
Mental Health in Tech

Team:
Byte & Brain

Colab link:

https://colab.research.google.com/drive/1BoBpiduhuc3U4U_LzN2RDk-oPANnB99Q?usp=sharing

Presented By :
Prakash Nihal

Stage 1: Can Workplace Conditions Predict Mental Health Treatment in Tech?

Hypothesis:

Employees with poor workplace support and a family history of mental illness are more likely to seek treatment.

- 01.** *Mental health is often ignored in the tech industry due to workplace stigma.*

- 02.** *Can we predict if a tech employee will seek mental health treatment based on their workplace and background?*

- 03.** *Employees with a family history of mental illness and poor workplace support are more likely to seek treatment.*

- 04.** *OSMI Mental Health in Tech Survey — 1,259 responses, 27 columns covering demographics and workplace conditions.*

- 05.** *<https://www.kaggle.com/datasets/osmi/mental-health-in-tech-survey>*

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Stage 2: Getting the data

Preparing the dataset:

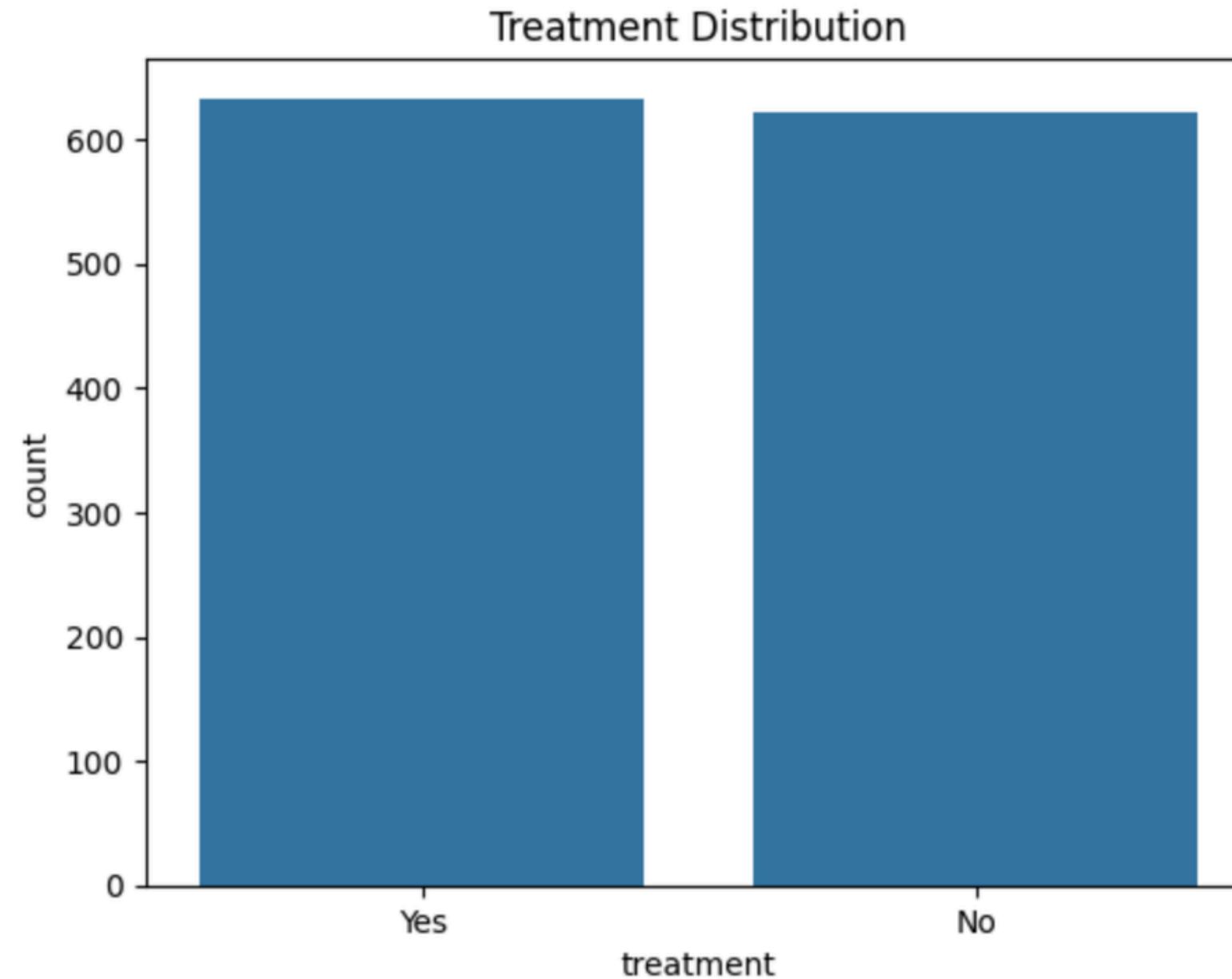
- Downloaded as a pre-made CSV file from Kaggle.
- Removed missing values, fixed inconsistent gender entries, and dropped irrelevant columns.
- Selected key features like age, company size, family history, and workplace support.
- Cleaned missing values
- Standardized gender entries
- Handled missing values using cleaning and preprocessing techniques

No additional datasets needed — the OSMI survey dataset is sufficient for the prediction.

Stage 3: Exploring the data

- 1,259 responses and 27 features
- Includes demographic and workplace-related variables
- Target variable: treatment (Yes/No)
- Most respondents are aged 25–35
- Data required cleaning due to missing values and outliers.

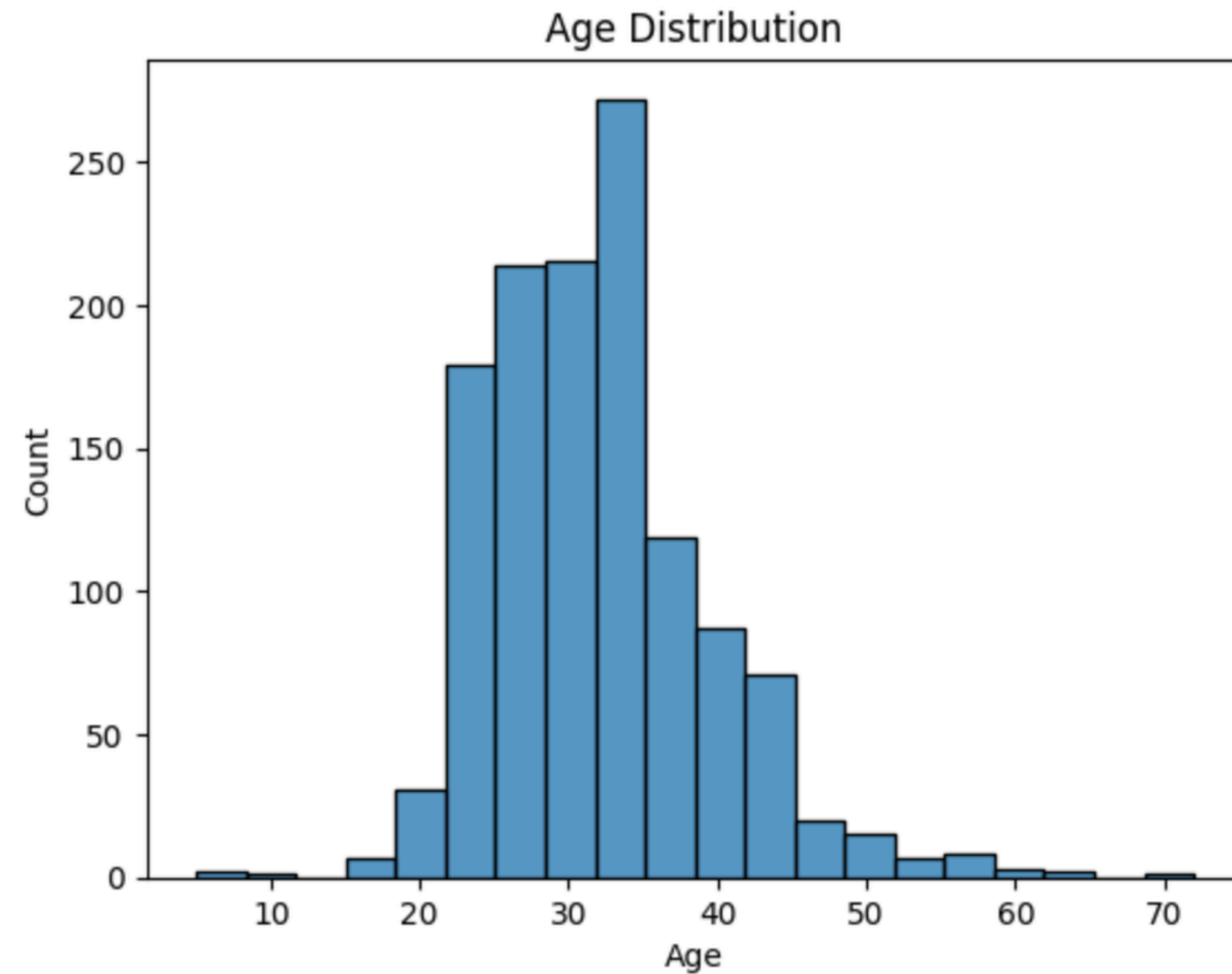
Treatment Distribution



- The distribution of employees seeking treatment versus not seeking treatment is relatively balanced, indicating that mental health treatment is common but not universal among respondents.

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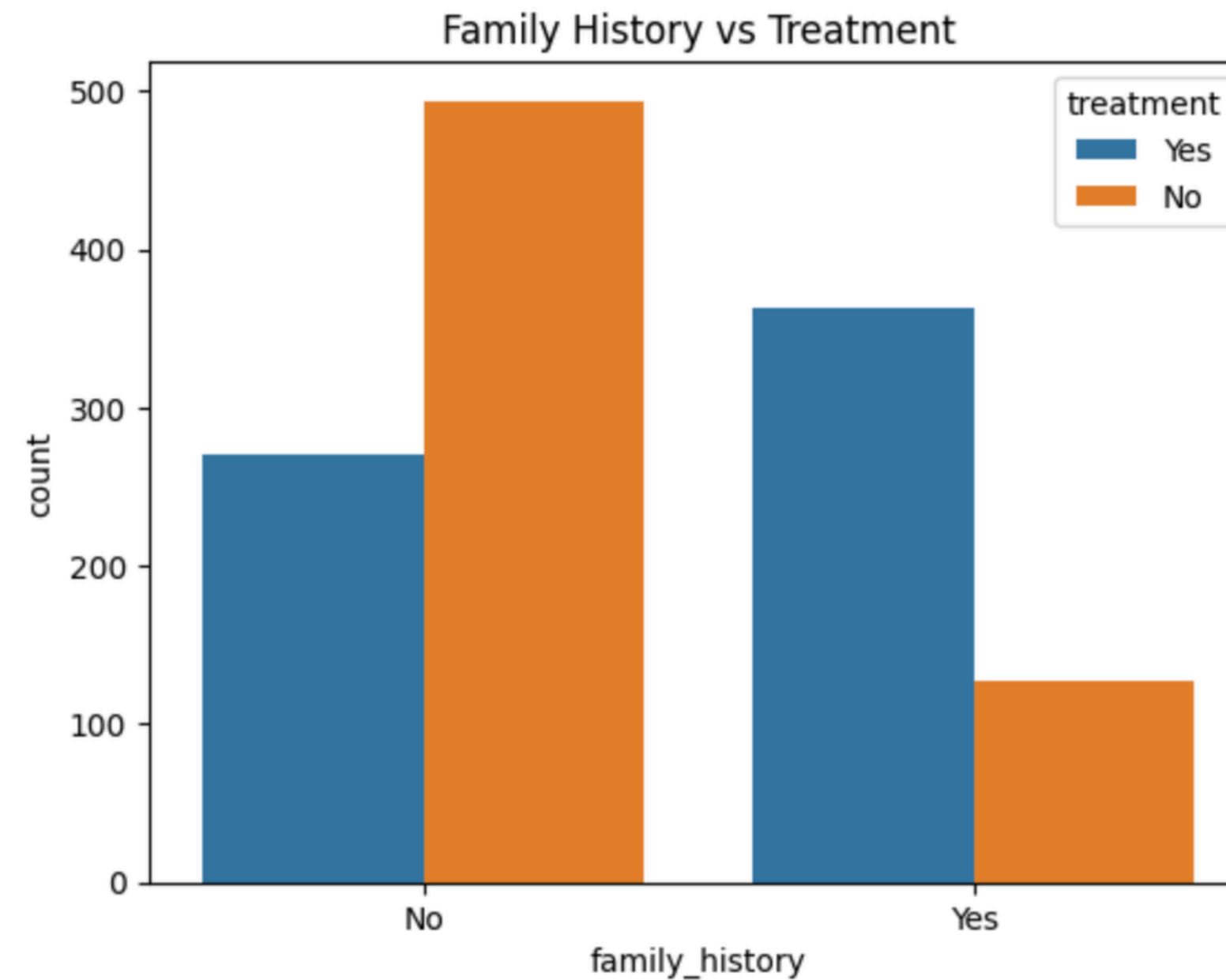
Age Distribution Graph:



- Most respondents fall between ages 25–35, indicating a younger workforce in tech.

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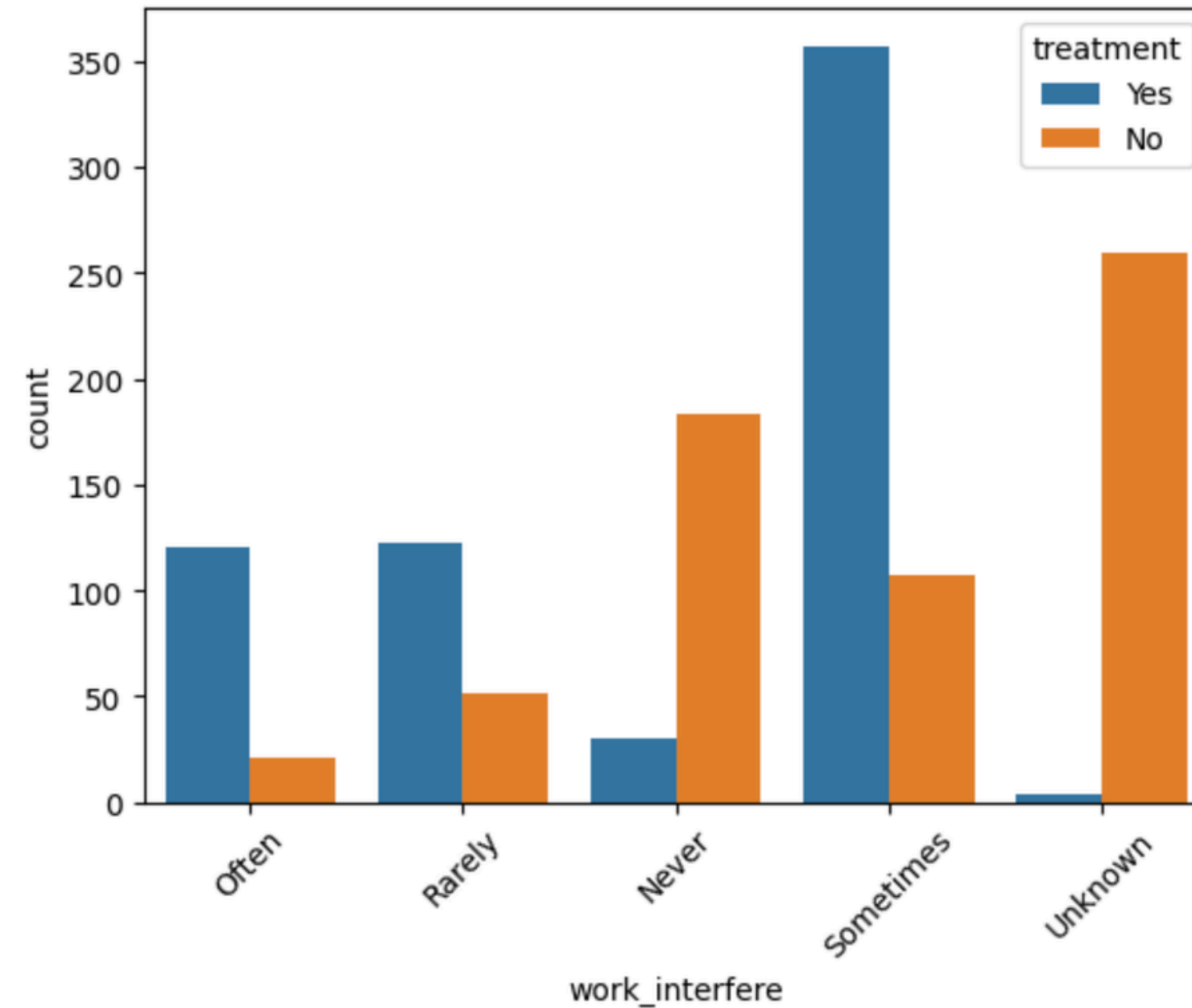
Family History vs Treatment



- Employees with a family history of mental illness are significantly more likely to seek treatment, making it a strong predictor.

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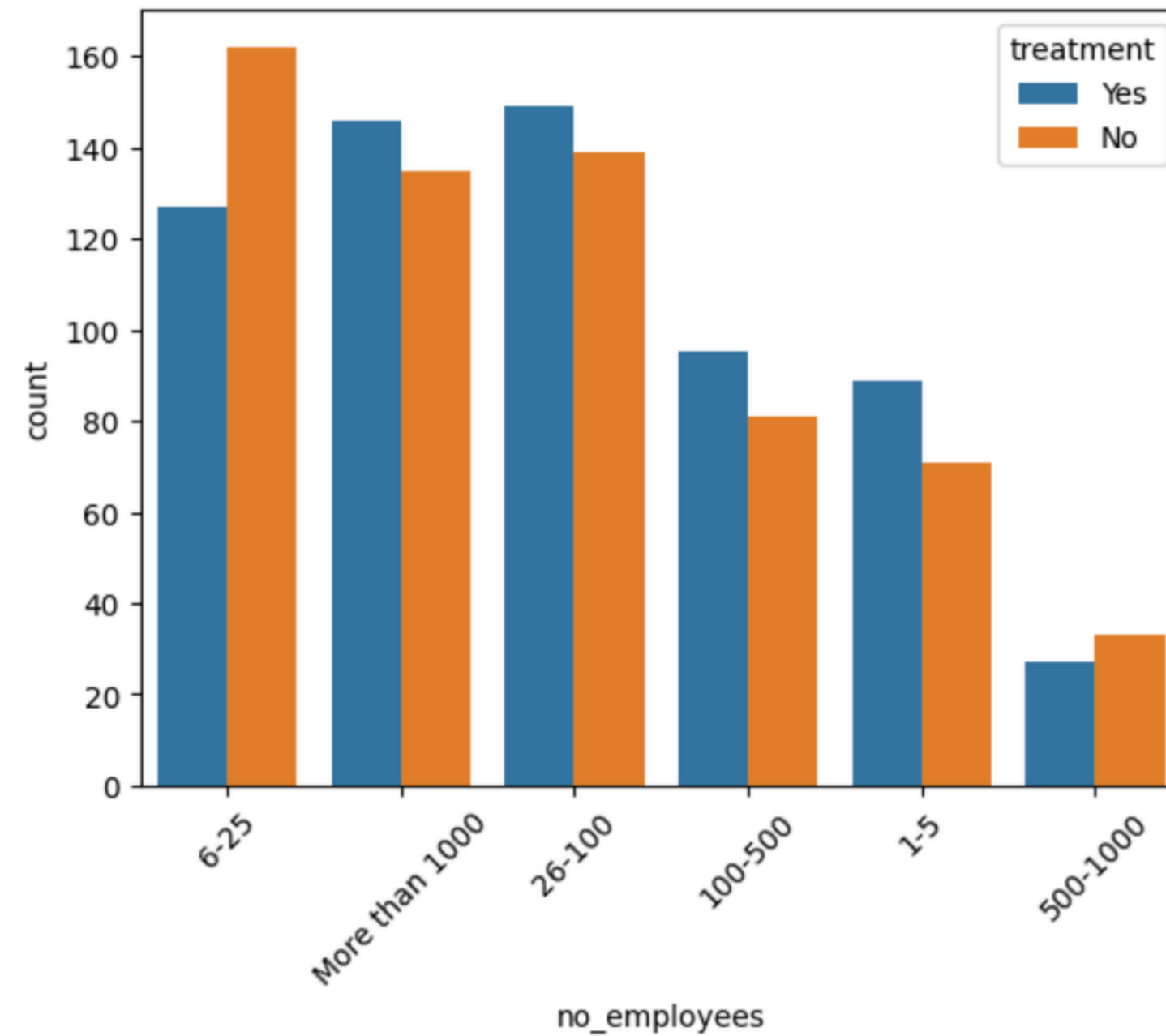
Work Interference vs Treatment



- Employees whose mental health interferes with work are more likely to seek treatment, highlighting the impact of workplace stress.

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Company Size vs Treatment



- Employees in larger companies tend to seek treatment more, possibly due to better access to mental health resources.

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Statistical analysis

Statistical Analysis Results

- Chi-square test shows a significant relationship between workplace factors and treatment
- Cramér's V indicates moderate association strength
- Confirms that variables like family history and work interference are important

Contingency Table:

```
Contingency Table:
treatment      No  Yes
family_history
No             494  270
Yes            127  363

Chi-Square Test Results:
Chi2 Statistic: 177.69583537275963
p-value: 1.5435847916844047e-40

Conclusion: There IS a significant relationship between family history and treatment.
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Cramér's V Analysis:

	Cramers V
work_interfere	0.686177
family_history	0.376435
benefits	0.227174
no_employees	0.083011
remote_work	0.025175

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Machine Learning Models

Predictive Modeling

- Logistic Regression model built for prediction
- Random Forest model used for comparison
- Dataset split: 80% training, 20% testing

Model Performance

Model Performance Comparison

- Logistic Regression Accuracy: ~81%
- Random Forest Accuracy: ~80%
- Both models performed similarly

This suggests relationships in the data are relatively simple and interpretable.

Feature Importance

Key Predictors (Model Insights)

- Family history (strongest predictor)
- Work interference
- Workplace benefits

These features consistently influenced predictions across models.

Key Findings:

- Family history is the strongest predictor of treatment
- Workplace conditions significantly influence mental health decisions
- Machine learning models confirm patterns observed in statistical analysis
- Simpler models perform effectively, indicating structured relationships in data

Limitations:

- Data is self-reported, which may introduce bias
- Dataset mainly represents tech workers, so results may not apply to other industries
- Some missing or inconsistent data required cleaning
- Limited demographic diversity in responses
- Correlation does not imply causation

Conclusion:

Both statistical analysis and machine learning models indicate that workplace conditions and personal background significantly influence mental health treatment behavior among tech employees.

The consistency between statistical tests and predictive models strengthens the reliability of these findings.

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